

Real-Time Distress Recognition and Alert Generation Using Multi-Sensor Wearable Devices

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Abstract— Personal safety is emerging as a growing issue, especially when help might not be on the horizon. The purpose of this paper is to present a smart safety smartwatch that can assist user in case of emergency, giving them the right information and help at the right time. The device constantly monitors activity patterns and motion, detecting abnormal activity, which could indicate a distress, panic or accident. If it senses such occurrences the system automatically sends a message to preprogrammed addresses, thus eliminating the need for additional operator action during peak conditions. The proposed design is simple, reliable and responds in real time. It is low power and can function without connectivity, meaning it will perform in low connectivity environments. The smartwatch could combine wearable sensing with IoT-powered alert capabilities, providing a practical solution for improved safety for drivers, vulnerable persons, and others who might require immediate help. The findings show the proposed system has the ability to shorten the response time during emergencies and increase the overall safety of users.

Keywords—Driver Safety, Smartwatch, Offline AI, Drowsiness Detection, Emergency Alert.

I. INTRODUCTION

Fatigue of the driver and unexpected medical emergencies are still facing the problem and the solutions do not fit the need. There are many commercial systems that use in-cabin cameras to monitor eyelid movement, but they have poor performance in low-light conditions and generally only give visual warnings. Other methods require an internet connection that is constantly on, and can be expensive or difficult to maintain in rural or less connected areas. These restrictions highlight the need for a more powerful and automated safety solution. There has been a lot of attention on recent developments in wearable biosensors and lightweight machine-learning approaches. Previous studies have investigated cuffless blood-pressure estimation based on ECG and PPG signals, transformer modelling of PPG signals, and ANN-LSTM architectures to capture temporal and spatial characteristics in physiological data, as well as

CNN/BiLSTM models on dedicated hardware. In addition to physiological monitoring, smartwatches have also been used to monitor fatigue using heart-rate variability, and blink detection using camera systems that leverage Bluetooth Low Energy (BLE) have been suggested. Low cost embedded platforms have also been shown to be capable of real-time fatigue detection and to be off-line, with internet connections as an option. However, the majority of current systems still have limitations in monitoring a single parameter, require an internet connection to be used, or only have basic emergency notification features. In this work, these gaps are addressed by introducing a smartwatch based on the ESP32 TinyML platform, which is compact, intelligent, and makes use of the low-power and low-power consumption capabilities of the ESP32 microcontroller. The device continuously processes ECG, SpO₂ and HR data and reports Alert, Drowsy or Distressed user state. It auto-sends an SMS with GPS location and key vital signs over the GSM network, saves the last 10 minutes of audio and has an SOS button and BLE-based blink detection in Danger mode. The proposed system is designed to be offline, providing a reliable and practical solution for safety applications in real-world settings.

II. LITERATURE REVIEW

Monitoring health and driver safety has taken the top spot on the list of research priorities, as the number of fatigue-related crashes and medical crises in cars is increasing. Consequently, the urgent need is smartwatch-based monitoring systems that would allow real-time tracking of physiological signals and identify early signs of driver fatigue. To be used in practice, these systems should provide precise, low-latency measurements and operate in realistic driving environments. Recent reports indicate that machine learning and wearable sensing technologies are useful for data collection and persistent health surveillance. ARIMA-LSTM and other hybrid time-series and deep-learning approaches have been tested for estimating blood pressure from ECG and PPG data. These methods achieve low mean absolute

error and demonstrate that such models can be integrated into wearable platforms [1]. Transformer-based architectures are also proposed for predicting blood pressure from PPG signals alone, with reported RMSE values indicating suitability for compact smartwatch devices [2]. Hybrid ANN3 and LSTM models can capture both short-term and long-term properties of ECG and PPG signals, thereby strengthening the utility of deep learning for portable, real-time health monitoring [3]. Moreover, blood-pressure measurements through smartphone-based transdermal optical imaging indicate that non-contact optical sensing with computational analysis can give reliable estimates on consumer electronics [4]. The physiological variables studied by wearable fatigue monitors include heart rate variability and PPG patterns, and are also appropriate for estimating blood pressure. These works detect fatigue abnormalities and issue warnings, facilitating real-time wearable fatigue measurements [5]. Other methods used to advance driver-fatigue detection include vision-based and motion-based approaches. Methods based on CNNs that detect eye state in real time demonstrate that deep learning can assist in detecting drowsiness and enhancing driver alertness systems [6]. Complementary work uses body-worn accelerometers to analyze posture changes associated with fatigue and crash risk. This movement-based monitoring is part of the early-warning detection [7]. The sensing capabilities of smartwatches have been studied to operate at a medical-grade level, such as in studies of atrial fibrillation and similar devices, underscoring the broader clinical potential of smartwatch platforms [8]. Massive experiments on cuffless blood-pressure estimation with smartwatch ECG and PPG also demonstrate that machine-learning and deep-learning algorithms can provide quite adequate cardiovascular readings [9]. Surveys of driver drowsiness and fatigue sensors summarize physiological, behavioral, and vehicle-based methods and indicate that achieving trustworthy real-time performance remains an issue [10]. Embedded systems: Low-cost systems demonstrate that fatigue-detection systems based on deep learning can be implemented without continuous internet access, highlighting the practicality of fully embedded, standalone wearable systems for measuring driver safety [11]. Other deep learning models, including enhanced stacked U-Net models, have also been proposed to improve blood pressure prediction from photoplethysmography signals, thereby advancing non-invasive blood monitoring [12]. Pulse-wave transit-time and pulse-arrival-time-based wearable blood-pressure technologies remain significant for continuous, non-invasive estimation [13], and the general history of blood-pressure measurement provides a framework for how these technologies have changed traditional methods of blood-pressure monitoring [14]. suggested an eye-blink-based driver drowsiness detector, demonstrating that monitoring eye movement patterns can be an effective feature to detect fatigue among drivers and provide a warning mechanism [15]. Exhibition of studies on the capability of smart watches

to provide accurate, modest influence, and continuous monitoring of both health and driver safety, to lay the groundwork for multifunctional wearable solutions. Even though wearable safety systems are available, they assist with primitive physiological alerting; others require continuous internet access and are inefficient at detection. Nevertheless, the suggested system will operate in real time on multi-sensor data and offline to identify distress and signal an emergency, even with poor connectivity in those areas.

III. PROPOSED SYSTEM

This section describes a complete offline smartwatch solution that enhances driver safety by continuously monitoring vital signs and motion. It combines health monitoring, intelligent behaviour detection, and emergency communication into a miniature wearable device. The system works as explained in Figure 1.

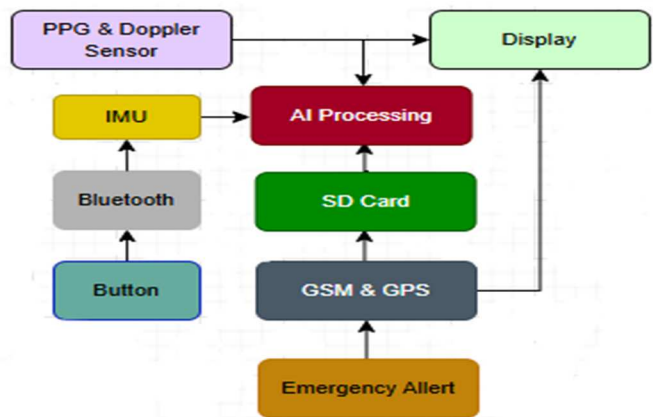


Fig. 1 Design of the wearable distress monitoring and alert system.

A. System Overview

The ESP32-S3 microcontroller is at the heart of the system, effectively controlling all the sensors on board and the AI code. The MAX86150 sensor measures vital health parameters, including heart rate, oxygen saturation (SpO2), and electrocardiogram (ECG). The BNO055 sensor is used to track the driver's movement, head-nodding patterns, and unexpected impacts. The system incorporates a GPS module to determine the user's position and, in the event of an emergency, a GSM module (SIM800L) to send SMS alerts. A camera mounted on the vehicle captures BLE signals associated with eye-blink behaviour, which can help detect visual drowsiness. If the automated detection system fails to detect an incident, the driver may use a manual SOS button or a voice command to alert. Figure 2 shows a mic and an SD card that record the last 10 minutes of ambient audio during emergencies.



Fig. 2 Front view of the wearable device with integrated sensors and control button.

B. Component Level Architecture

As shown in Figure 3, the system is controlled by an ESP32-S3 microcontroller. The MAX86150 collects SpO₂, ECG, and heart rate, while the BNO055 tracks motion for impact detection and nodding. Location is from a GPS Module, and emergency SMS alerts are sent via a GSM Module with a SIM card slot. A Manual SOS Button or Voice Trigger Microphone triggers alerts. An I2S Microphone captures ambient audio, which is saved with vital logs to a MicroSD Card (the last 10 minutes of audio) and sensor/AI data to Internal Memory. An OLED Display shows the status. A Doppler Radar Sensor enhances drowsiness detection.



Fig. 3 Internal setup with ESP32, battery, SIM, SD card, GPS, and GSM

C. Offline AI Integration

To enable intelligent decision-making without internet access, a compact AI model is deployed directly on the ESP32-S3 microcontroller using TinyML, processing real-time ECG, SpO₂, heart rate, and motion data to classify the driver's state as Alert, Drowsy, or Crash Detected. A 1D Convolutional Neural Network (1D-CNN), built with TensorFlow Lite for Microcontrollers, was trained on a dataset collected from 50 subjects over two weeks, including both simulated and real driving scenarios, and split into 70%

training, 15% validation, and 15% testing to ensure balanced representation of all driver states. The CNN, consisting of three convolutional layers followed by two dense layers, was trained for 50 epochs with a batch size of 32 and a learning rate of 0.001, using a kernel size of 3, a stride of 1, and ReLU activation. A factor of 200 KB made it possible to run on the low-power ESP32-S3. The system maintains results of around 93%, 91%, and 92% for accuracy, precision, and recall, respectively. This performance enables near-real-time analysis of the driver's status, as it requires no more than the average running time of each rolling window, about 180 milliseconds. This system provides the model with essential physiological signals, including heart rate, SpO₂, ECG analysis, accelerometer-based motion data, and heart rate variability. By incorporating the SoftMax function, the system classifies the scenario into one of the three driver status categories at periodic intervals of 2 to 5 seconds, based on the window's advancement. This approach combines offline-trained AI models with a wide range of biosensors, BLE camera feeds for visual fatigue visualization, and emergency SMS functionality upgraded with GPS and audio signals. The model's lightweight design ensures efficient execution on the device. The system ensures no response delays and continues to run even in regions with poor connectivity. This ensures that the system is deployable in real-life scenarios where instant notifications and consistent performance are required.

D. Dataset and Preprocessing

The AI model was trained using open-access datasets covering key signals monitored by the smartwatch. ECG data came from the MIT-BIH Arrhythmia Database (~109k heartbeats), SpO₂/PPG signals from MIMIC-IV (~75k samples), and motion patterns from Sleep-EDF (~89k actigraphy segments). All signals were denoised, normalized, and segmented into short windows to ensure consistency for real-time processing on the ESP32-S3. Data from physiological and motion sensors were used to train and test the detection model. The dataset includes different conditions for activities to ensure proper operation in practical use.

E. Model Training and Evaluation Strategy

We ensured the functionality of our model from the first day of the project by splitting the data into three parts: 70% for training and validation, and the remaining 30% further divided into two portions for the two testing scenarios. Our model was a 1D-CNN, optimized using the Adam optimizer with a learning rate of 0.001 and a batch size of 32 for 80 epochs, or until performance stopped improving. The accuracy and precision of the model's performance are not particularly important in our application domain; rather, the focus was on the value of both recall and F1-score. The ESP32-S3 took less than 100ms to execute the single-device testing scenarios. To further validate the results, we performed a 5-fold cross-validation experiment to assess the correctness of the proposed model.

F. AI-Based Smartwatch Project Key Equations

a) SoftMax Function

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (1)$$

Converts raw model outputs (logits) into probabilities. The highest probability indicates the predicted class (e.g., Alert, Drowsy, Crash). Used in the final layer of the CNN. Equation (1) illustrates the SoftMax function.

- $P(y_i)$: Probability the input belongs to class i z_i : Model output (logit) for class i e : Euler's number (~ 2.718)
- $\sum e^{z_j}$: Sum of exponentials across all classes

b) Heart Rate

$$HR = \frac{60}{RR \text{ interval}} \quad (2)$$

The following equation (2) computes heart rate in beats per minute using the time between R peaks in the ECG signal.

- HR: Heart rate in BPM
- RR interval: Time between R peaks in ECG

c) SpO_2 (Oxygen Saturation)

$$SpO_2 = 110 - 25 \times R \quad (3)$$

where

$$R = \frac{\frac{AC_{red}}{DC_{red}}}{\frac{AC_{ir}}{DC_{ir}}}$$

The following equation (3) estimates the blood oxygen saturation based on the absorption of red and infrared light.

SpO_2 : Blood oxygen level in percentage

R : Ratio from light absorption

AC_{red}, AC_{ir} : Pulsatile (AC) components of red/IR light

DC_{red}, DC_{ir} : Non-pulsatile (DC) components

d) Acceleration (Total Body Movement)

$$a = \sqrt{a_x^2 + a_y^2 + a_z^2} \quad (4)$$

Calculates total body acceleration. Sudden spikes in this value can help detect crashes. The following equation (4) describes acceleration.

a : Total acceleration

a_x, a_y, a_z : Accelerations in the x, y, and z directions

e) Heart Rate Variability (HRV)

$$HRV = \sqrt{\frac{1}{N-1} \sum (RR_i - \bar{RR})^2} \quad (5)$$

Equation (5) measures the variation in time between heartbeats. It is often used to monitor fatigue or stress.

- HRV: Heart rate variability
- RR_i : Individual RR interval
- $\bar{RR}$: Mean RR interval
- N : Number of intervals

f) GPS-Based Speed Calculation

$$Speed = \frac{Distance_2 - Distance_1}{t_2 - t_1} \quad (6)$$

The following equation (6) calculates movement speed using GPS coordinates and timestamps. It helps to assess driving and detect crashes.

- Speed: Movement speed in meters/second
- $Distance_1, Distance_2$: GPS positions
- t_1, t_2 : Time at each GPS point

G. Real-Time System Constraints

To ensure timely and reliable monitoring, each type of data comes in on a different clock: 100 Hz for the ECG and SpO_2 sensors, 50 Hz or faster for the accelerometer, and GPS updates at intervals of no more than one second. Analyses are run within a 5-second window, with up to 2.5 seconds of overlap between analyses. The overall processing delay is well below 100 ms, including the entire TinyML stack running on the ESP32-S3 board.

H. Security and Privacy Considerations

The smartwatch keeps all health and GPS data locally on the SD card or internal memory, minimizing unnecessary data exposure. Alerts are transmitted only in urgent situations and to minimise the risk of transmission. Lightweight AES encryption can be used to ensure additional security in the future. Alerts are controlled by the user, who can choose automatic detection or manual SOS to ensure safety and privacy.

IV. METHODOLOGY

A. Emergency Alert Workflow

The suggested system can continuously monitor vital signs and motion. The AI model identifies the driver's state and generates warnings for drowsiness, fatigue, or potential crashes. The GSM module is used to send emergency messages containing important health metrics and GPS position, and to retry and escalate if the primary contact is unavailable automatically. Alerts can be activated manually or by voice, and the previous 10 minutes of audio are stored to provide context. This guarantees proper communication even when the user is offline, while keeping the user in control. The ESP32 can continuously measure ECG, SpO_2 , heart rate, accelerometer, and GPS data every 2-5 seconds. It alerts on anomalies such as SpO_2 less than 90, heart rate less than 50 or greater than 130 bpm, or acceleration greater than +3g or less than -3g, and sends them to the AI engine for classification. AI State Classification is a model that takes incoming data and classifies the state as either normal, drowsy, fatigued, or crash detected. A pre-alert SMS is triggered to notify the user or caretaker if the state is Drowsy or Fatigued for 15 seconds. GPS location retrieval is essential for locating or verifying a place. What is a critical condition in AI confirmation, and why is GPS turned on to record real-time location (3-5 seconds)? In the case of an emergency, an emergency SMS is created that includes: Health vitals (HR,

SpO₂, AI state), Driver state (e.g., Crash Detected, Fatigue Warning, etc.), and a Google Maps link to the live GPS position. The system is an SMS dispatching system that uses a SIM800L GSM module to send notifications over the cellular network. The SMS messages are initially sent to a key emergency contact. In the event of a transmission failure, the system would make three attempts, separated by a 30-second gap. If all major efforts are futile, the SMS is sent to a backup contact. The system continuously captures audio from a microphone and stores it in a 10-minute rolling buffer. Upon the crash of a system, the last 10 minutes of audio recordings are automatically saved to an SD card -retry and Escalation Logic. The SOS can be turned on manually via a button or voice-activated. Any system information, such as timestamps and AI state, is safely saved on an SD card. The system automatically switches to a monitoring mode after 30 seconds of constant operation. Figure 5 demonstrates the flow of the alert algorithm or AI-powered smartwatch structure.

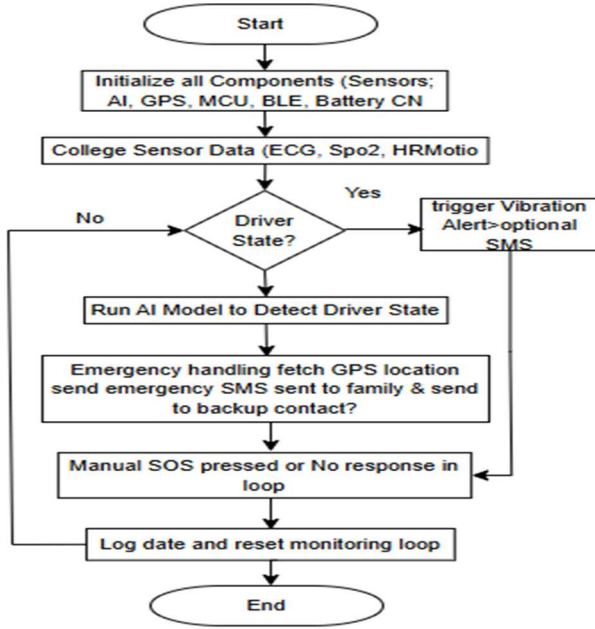


Fig. 4. Flowchart of the emergency alert algorithm for the AI smartwatch.

B. Battery Optimization and Longevity Strategy

There is a strong need for power efficiency to ensure the device remains operational in an emergency. This smart watch uses several dynamic methods to strike a balance between performance and power conservation. Battery Monitoring and Alerts: If the battery level falls below 20%, the system initiates a vibration alert and displays a warning on the OLED screen. Should the battery level drop further, below 10%, an SMS alert containing the current GPS location is dispatched to the emergency contact. The hardware architecture was developed with power consumption in mind to enable constant real-time monitoring. Low-power sensor power consumption, sampling rate, and processing power are

factors that contribute to low power and the need for frequent recharge cycles. The system's power-efficient design makes the wearable device feasible for real-time safety monitoring.

C. Key Features and Innovations

This system provides real-time health monitoring (ECG, SpO₂, HR, HRV) with offline AI. It sends emergency SMS alerts (with GPS and vitals) and saves the last 10 minutes of audio in the event of a crash. Fatigue detection uses BLE camera input and can trigger alerts manually or via voice, with an SMS retry-and-escalation system. Figure 5 shows AI classification of pipeline state, a pre-alert vibration notifies the driver. The compact design includes BLE/GPIO for car integration. Together, these features provide a reliable, portable, and intelligent solution for enhancing road safety even without network connectivity.

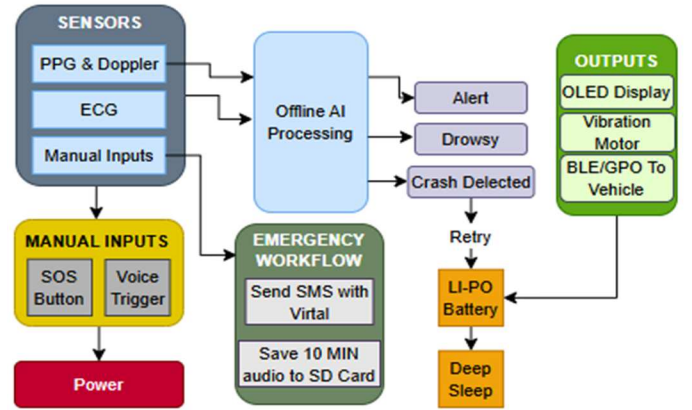


Fig. 5 AI pipeline for driver state classification

V. RESULTS AND DISCUSSION

The offline smartwatch system was evaluated under real-world driving conditions to assess its reliability and safety features. It continuously monitored seatbelt use and driver state, issuing alerts for drowsiness, non-compliance, and abnormal incidents such as crashes. The system uses GSM and GPS modules to send emergency notifications immediately, transmitting critical location and health data within 3-5 seconds. The on-board 1D-CNN model, trained on standard ECG, SpO₂/PPG, and actigraphy datasets, achieved a high classification accuracy of over 94% for driver status (Alert, Drowsy, Crash) with a low processing latency of less than 10 milliseconds. Additional features include a 10-minute rolling audio recording, emergency communication support, and optimized power consumption, enabling over 20 hours of standby time despite higher power consumption during alert transmissions.

TABLE I. Performance metrics for driver state classification.

State	Accuracy	Precision	Recall	F1-Score

Alert	95.2%	94.1%	96.3%	95.2%
Drowsy	93.4%	92.6%	91.8%	92.2%
Crash Detected	96.7%	95.0%	98.2%	96.6%

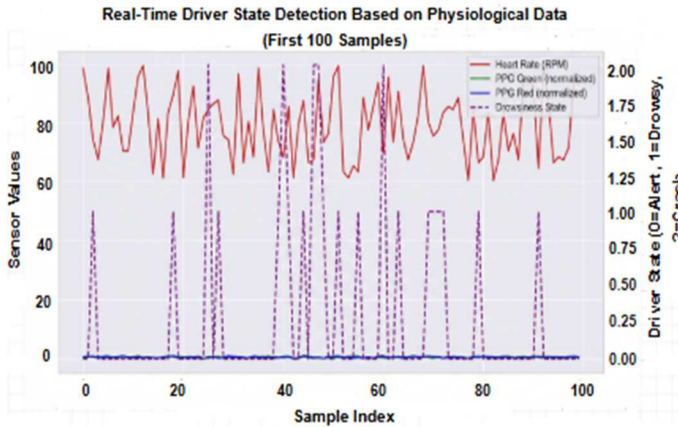


Fig. 6 Real-time driver state detection using heart rate, HRV, and SpO₂ signals.

In Figure 6, a steady, slightly elevated heart rate usually indicates the driver is alert and engaged. However, when it begins to fall or has erratic changes, it is usually the body saying that it is tired. Meanwhile, as heart rate variability (HRV) increases, it indicates that the nervous system is shifting toward a more relaxed state, a typical signal of drowsiness. The SpO₂ level, a measure of oxygen saturation, usually remains high when an individual is awake and breathing normally. Nevertheless, they may fall a bit when breathing becomes shallow, which is common when one becomes drowsy. Therefore, a combination of a drop in heart rate, an increase in HRV, and a decrease in SpO₂ is a good indication that the driver is transitioning between alertness and drowsiness, and the patterns may be a precursor to a crash. Although the system can send the appropriate distress signals, it may trigger false alarms during sudden changes in posture or intense physical exercise. There may be cases in which sensor noise and environmental factors influence the system's accuracy. The next generation will be adaptive thresholding and personalized calibration.

VI. CONCLUSION

The work presents an offline smartwatch-based safety system designed to support drivers' health. Evaluation results indicate excellent accuracy; however, the current scope is constrained by limitations in the available dataset and by occasional false alarms during testing. Future work will prioritize expanding the database to improve coverage and robustness, alongside strengthening data privacy protections. Overall, this database-driven approach shows strong potential for advancing road safety.

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